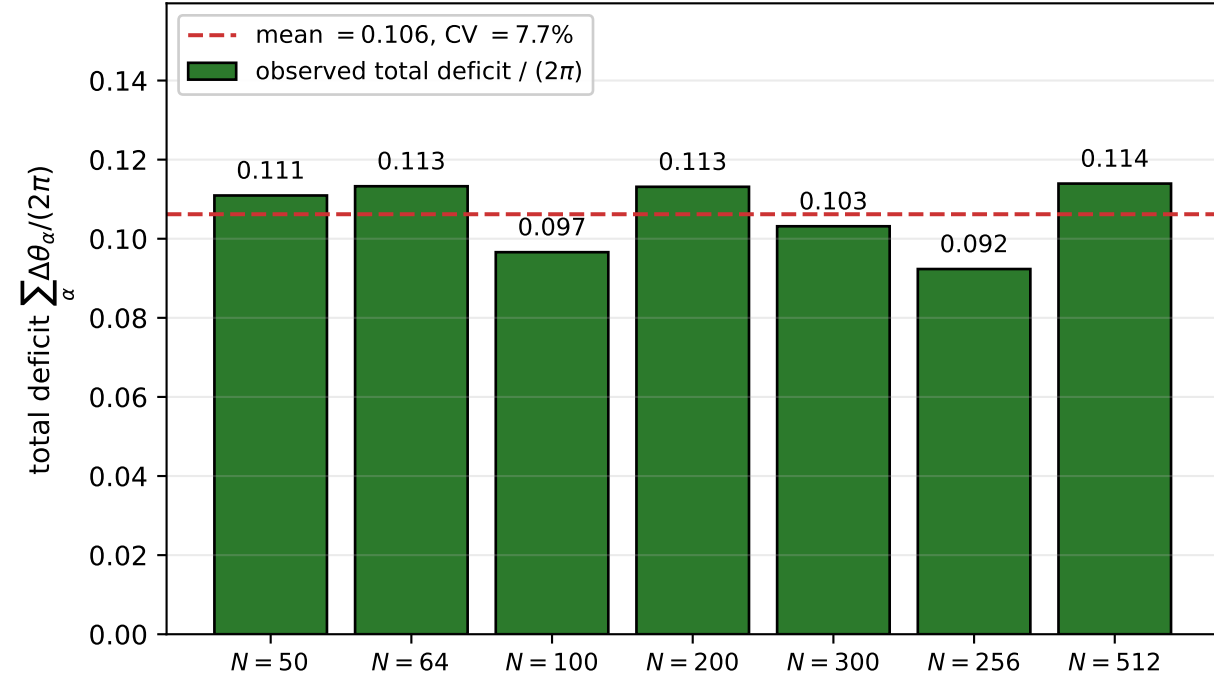
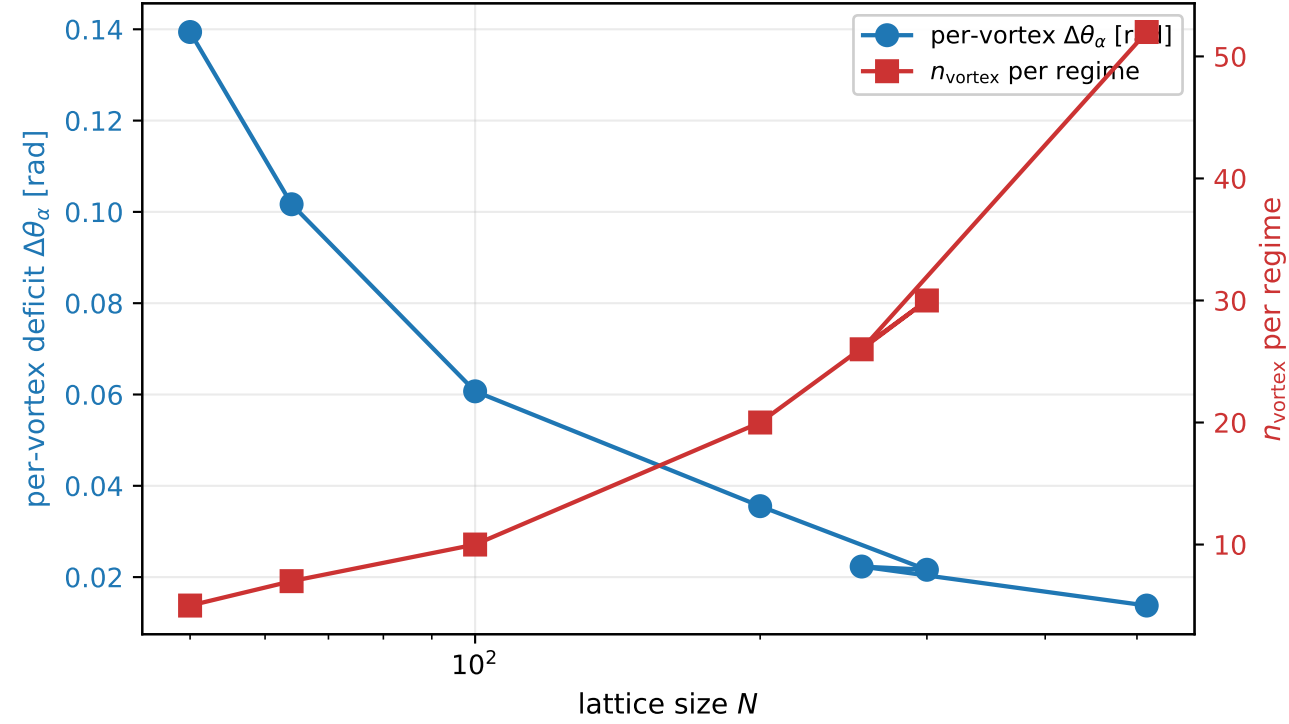


Quantitative defect-side analysis of the M3 sup-norm slack: topological invariant + matter-vortex co-localisation

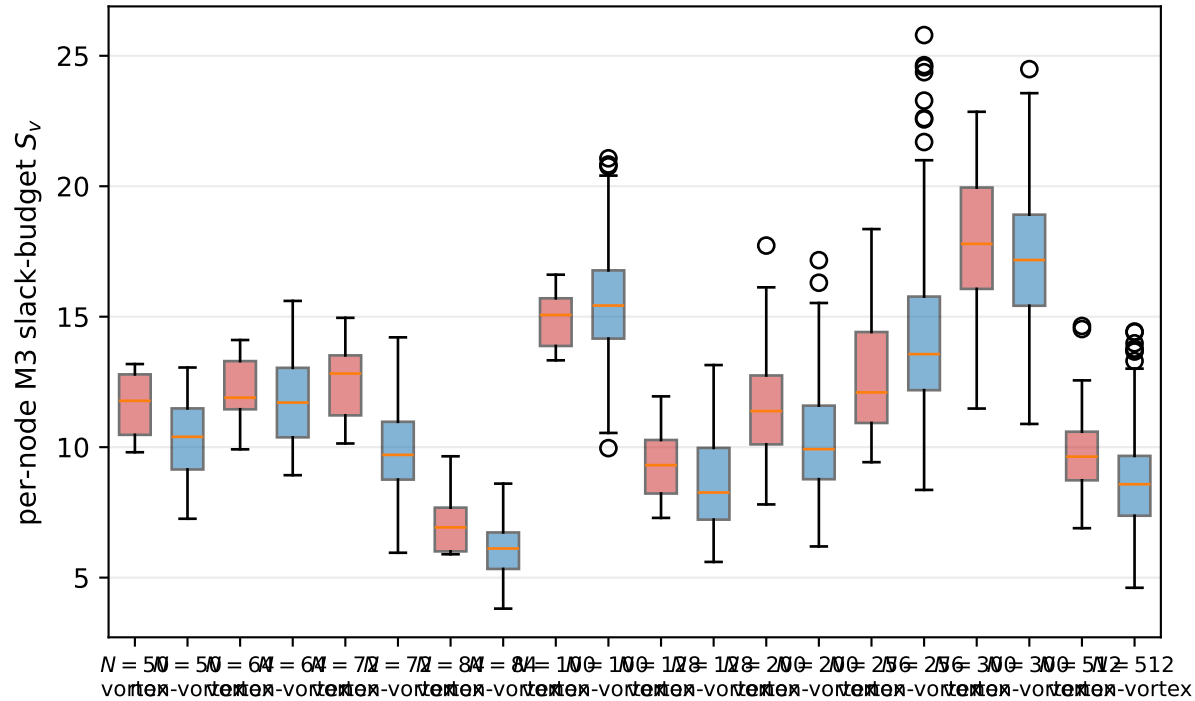
(A) Topological invariant $\sum_{\alpha} \Delta\theta_{\alpha}/(2\pi)$
on the within- P_5 ladder $N = 50 \dots 300$



(B) Per-vortex deficit decreases as n_{vortex} grows;
product is the conserved invariant of panel (A)



(C) Per-node slack-budget distribution by class:
vortex (red) vs non-vortex (blue) per N



(D) M3-slack-budget rank vs T_{00}^2 rank at P5N200: Spearman $\rho = 0.635$, $p = 5.2e - 24$
vortex nodes track the high-rank diagonal (matter-vortex co-localisation)

